CHAPTER 1: CLASSICAL MECHANICS & LAWS OF MOTION

1. Newton's First Law of Motion (Law of Inertia):
An object remains at rest or continues in uniform motion in a straight line unless acted upon by an external net fo

2. Newton's Second Law of Motion:
The acceleration of an object is directly proportional to the net force acting upon it and inversely proportional to it

3. Newton's Third Law of Motion:
For every action, there is an equal and opposite reaction. When object A exerts a force on object B, object B sim

CHAPTER 2: PRINCIPLES OF THERMODYNAMICS

1. The Zeroth Law of Thermodynamics:
If two thermodynamic systems are each in thermal equilibrium with a third system, they are also in thermal equili

2. The First Law of Thermodynamics:
Energy cannot be created or destroyed, only transformed from one form to another. The change in internal energ

3. The Second Law of Thermodynamics and Entropy:
Total entropy of an isolated system always increases over time. Heat cannot spontaneously flow from a colder b